Phishing Attack Simulation using MedusaPhisher

Introduction

In this project, a controlled phishing simulation was conducted to demonstrate how attackers create fraudulent login pages to steal user credentials. The objective was to understand the mechanics of phishing attacks, credential harvesting techniques, and how such threats can be detected and mitigated. The simulation was carried out strictly in a controlled and authorized environment for educational purposes.

Objectives

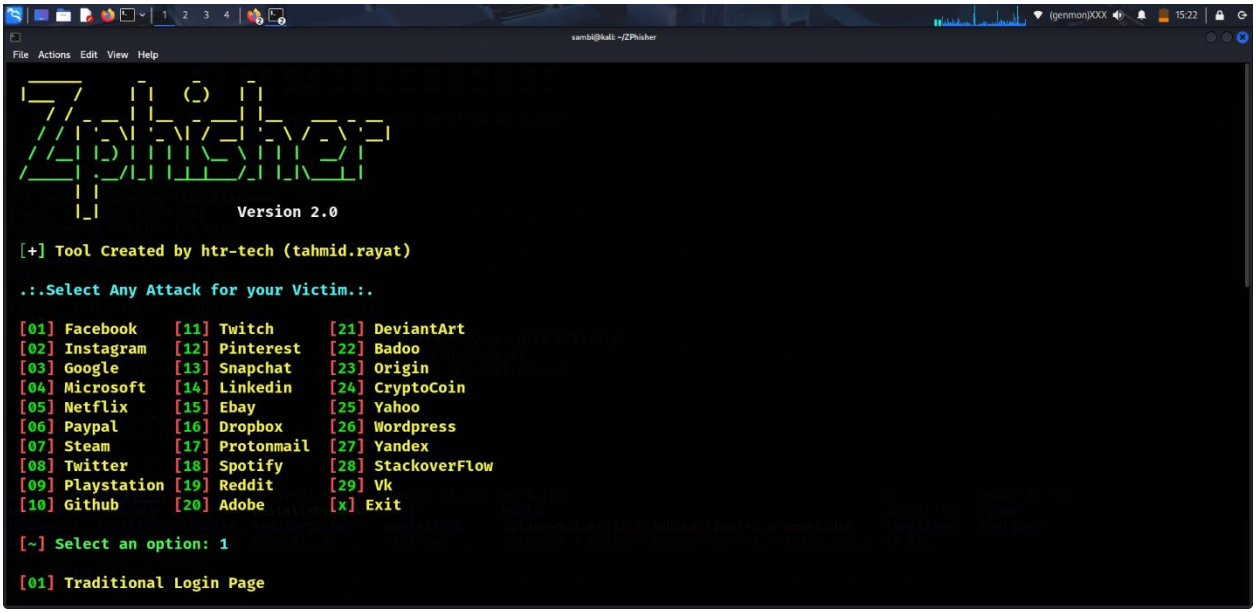
- To understand how phishing attacks are designed and executed.
- To simulate credential harvesting using a phishing tool.
- To evaluate user vulnerability to phishing attacks.
- To highlight effective mitigation strategies.

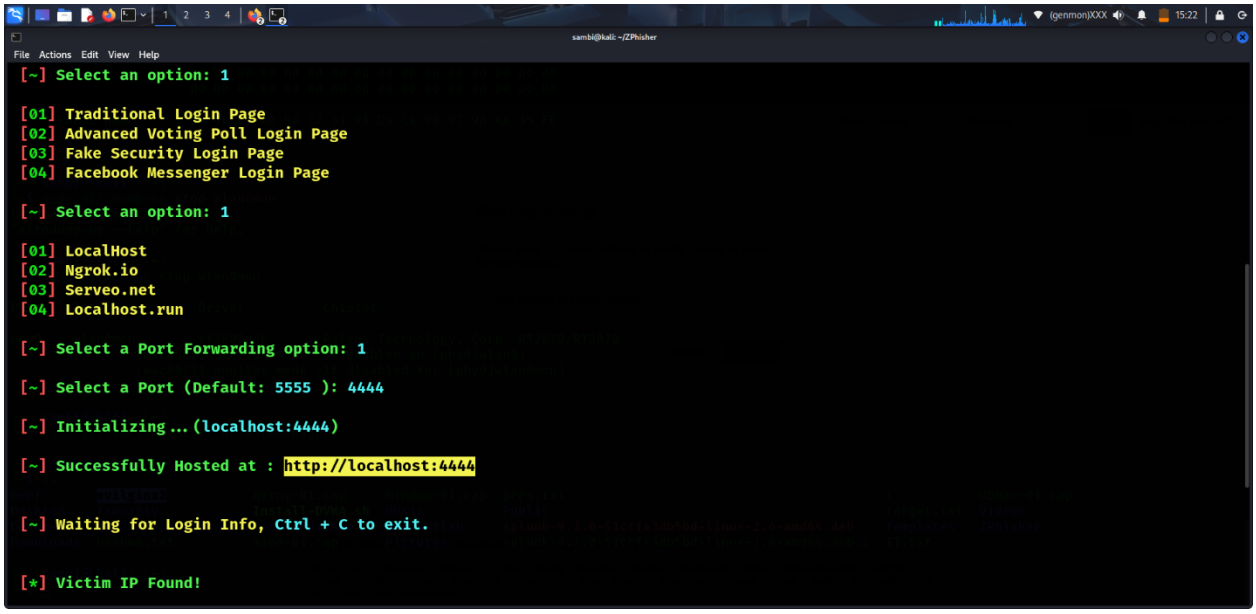
Tools and Environment

- ✓ Tool: MedusaPhisher / Zphisher
- ✓ Platform: Linux Environment
- ✓ Target Simulation: Facebook login page clone
- ✓ Network: Controlled lab setup

Methodology

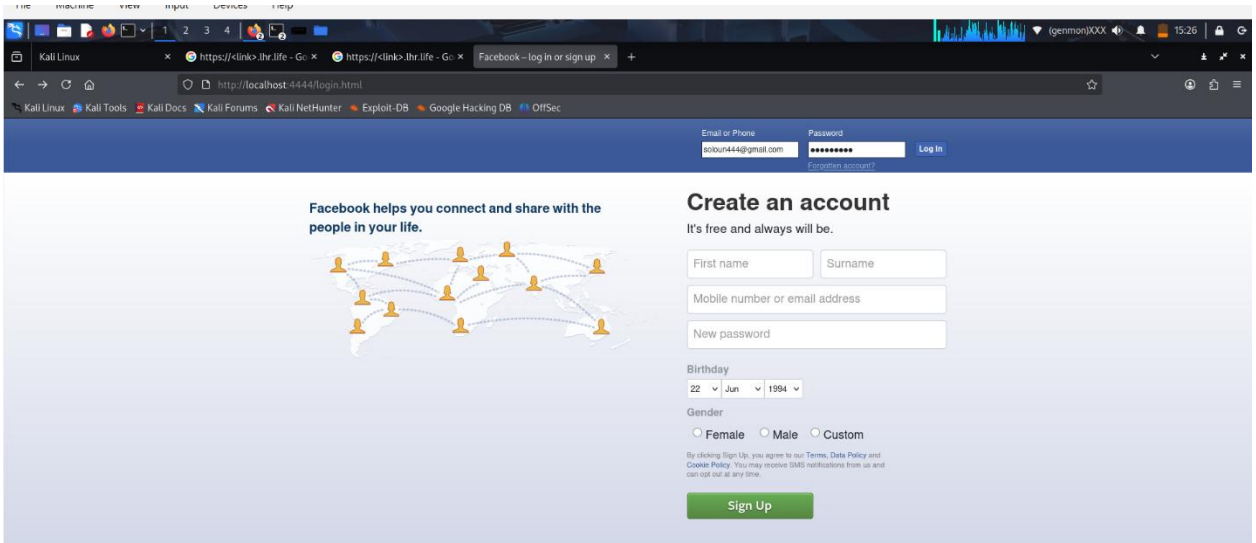
1. A phishing tool (Zphisher) was launched in a secure environment.





2. The Zphisher was set to face facebook page

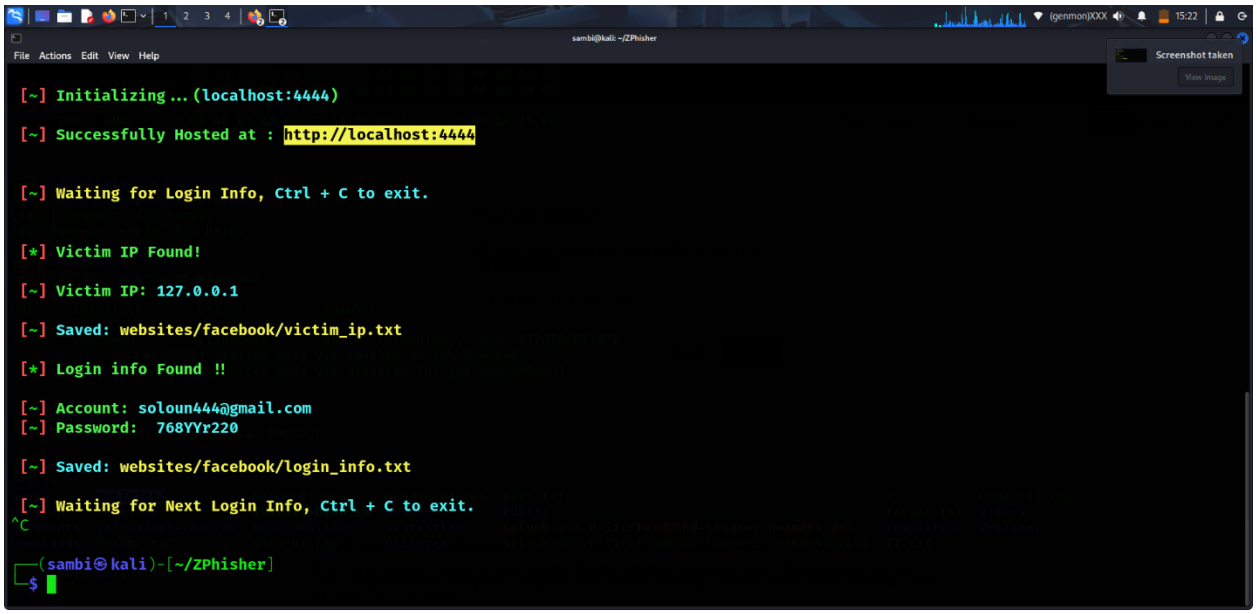
3. A replica of a Facebook login page was generated.



4. A phishing kink was created for the simulation purposes

5. Test credentials were entered into the fake login page.

6. The tool captured and displayed the credentials on the terminal.



Findings

The simulation demonstrated that phishing attacks can be highly effective, especially when the fake interface closely resembles a legitimate platform. Users without proper awareness can easily submit sensitive information, making credential theft a major cybersecurity risk.

Security Implications

- Phishing attacks exploit human trust rather than technical vulnerabilities.
- Weak user awareness increases susceptibility to attacks.
- Credential reuse across platforms amplifies damage.
- Lack of multi-factor authentication (MFA) increases risk.

Recommendations

- Implement multi-factor authentication (MFA) across all accounts.
- Conduct regular security awareness training for users.
- Educate users on identifying suspicious links and URLs.
- Use email filtering and anti-phishing tools.
- Encourage strong and unique password usage.

Conclusion

This project highlights the effectiveness of phishing attacks and the risks they pose to individuals and organizations. By simulating the attack in a controlled environment, valuable insights were gained into attacker techniques and user vulnerabilities. The

exercise reinforces the importance of cybersecurity awareness, strong authentication mechanisms, and proactive defense strategies.